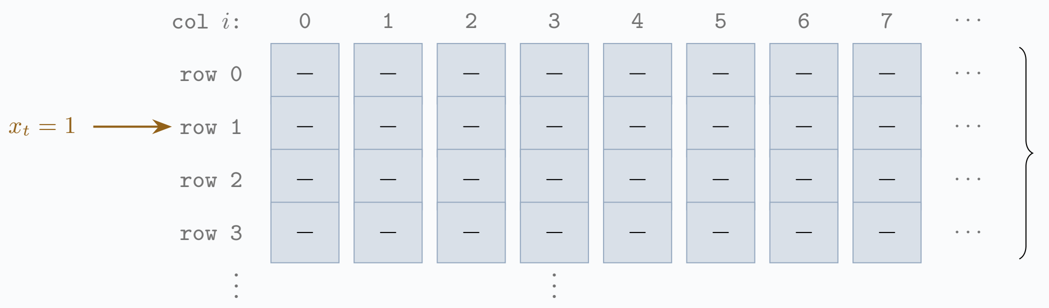
HDC-RWKV (nb12c) — Full Architecture Expansion

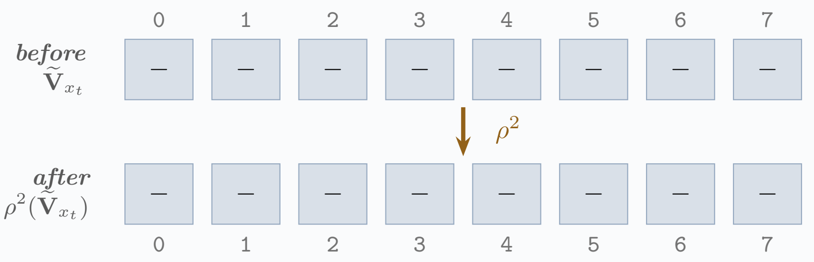
V=256, d=256, L=1, block size 16 • 16,434 deployment bytes • every learnable parameter bipolar at inference

Block 1. Token lookup — bipolar embedding table



$\tilde{\mathbf{V}} \in \{-1,+1\}^{V\times d}$
V=256 rows

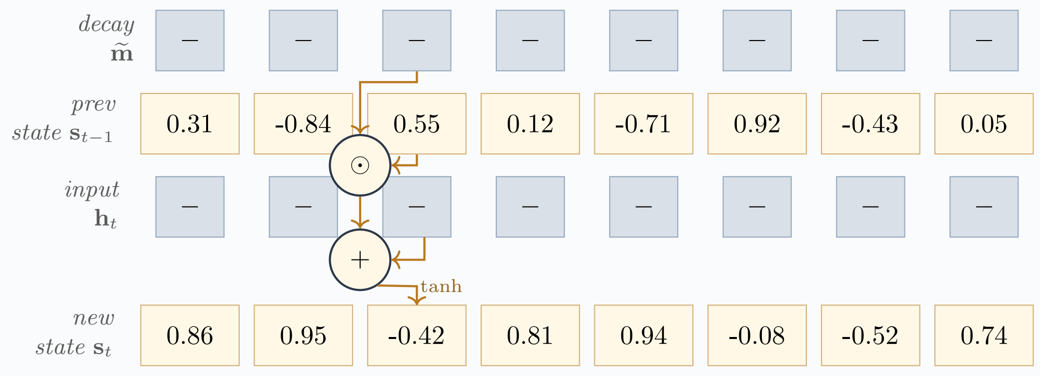
Block 2. Position binding ρ^t (cyclic right-rotation)



Cell at index j takes value from index $(j-t) \bmod d$.
Different positions produce orthogonal versions of the same token.

$\rho^t(\mathbf{h})_j = \mathbf{h}_{(j-t) \bmod d}$
(parameter-free, preserves $\{-1,+1\}^d$)

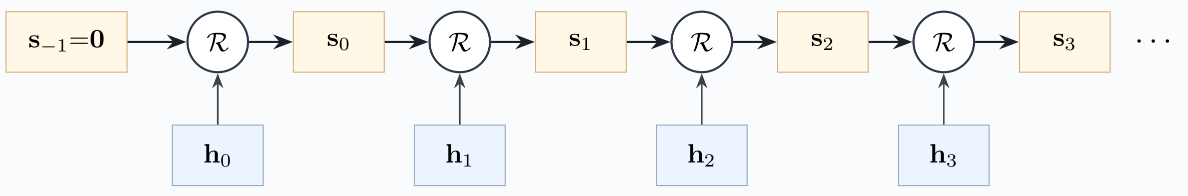
Block 3a. Recurrence — per-channel detail at one timestep t



channel i=2:
$\tilde{m}_2 s_{t-1,2} + h_{t,2}$
$= (+1)(0.55) + (-1)$
$= -0.45$
$\tanh(-0.45) \approx -0.42$

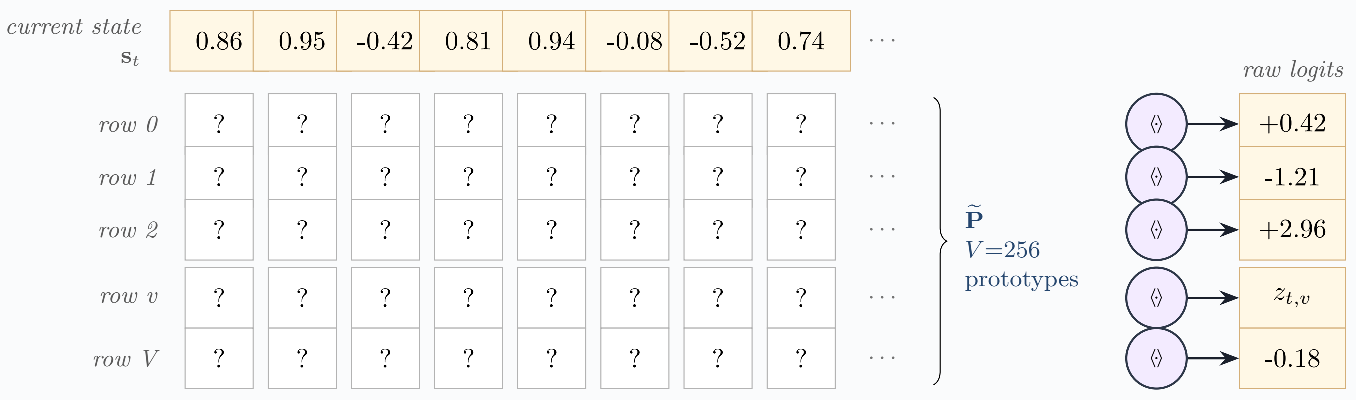
$\mathbf{s}_t = \tanh(\tilde{\mathbf{m}} \odot \mathbf{s}_{t-1} + \mathbf{h}_t)$ *(one channel at a time)*

Block 3b. Recurrence — unrolled across time

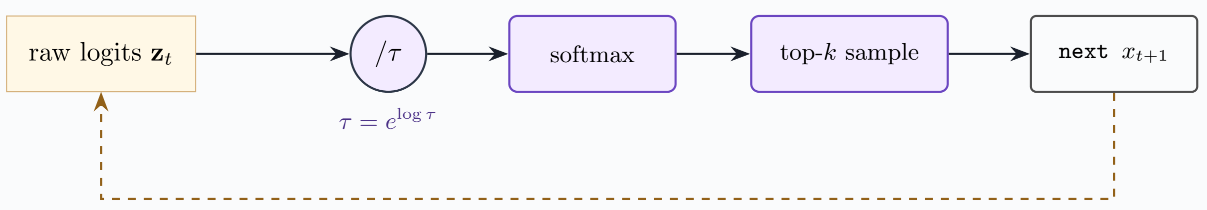


$\mathcal{R}(\mathbf{s}_{t-1}, \mathbf{h}_t) = \tanh(\tilde{\mathbf{m}} \odot \mathbf{s}_{t-1} + \mathbf{h}_t)$ *$O(d)$ per step. No attention. No key–value cache.*

Block 4a. Prototype-similarity output — one dot product per vocabulary item



Block 4b. Temperature scaling, softmax, top-k sampling, autoregressive feedback



autoregressive feedback: *next token becomes input at step t+1*

Aux A. Straight-Through Estimator ($\widetilde{\text{sign}}$)

Forward: $y = \text{sign}(x)$

Backward: $\partial y/\partial x = \mathbf{1}[|x| \leq 1]$

Forward is hard binarisation; backward is identity on the active strip $[-1,1]$ and zero outside. The continuous shadow $\mathbf{V}$ is updated by backpropagation; the recurrence always sees $\tilde{\mathbf{V}} \in \{-1,+1\}^{V\times d}$.

Aux B. Deployment equivalent — XOR + popcount

Encode $-1 \mapsto 0$, $+1 \mapsto 1$. Then for $a, b \in \{-1,+1\}^d$:

$$\langle a, b\rangle = d - 2 \cdot \text{popcount}(\bar{a} \oplus \bar{b}).$$

$\bar{a}$:	1	0	1	1	0	1	0	1
$\bar{b}$:	1	1	1	0	0	1	1	0
$\bar{a} \oplus \bar{b}$:	0	1	0	1	0	0	1	1

popcount = 4
$d - 2 \cdot 4 = 0$
(orthogonal)

The 6502 inner loop: XOR two d/8-byte rows, popcount the result, subtract 2·popcount from d. No multiplications, no floating point, no NaN reachable. For d=256: 32 bytes XORed and one popcount lookup per byte.